Fitoterapia 186 (2025) 106800

Contents lists available at ScienceDirect

Fitoterapia

journal homepage: www.elsevier.com/locate/fitote

Resveratrol in anti-biofilm therapy: Mechanisms, molecular derivatives, and emerging drug delivery strategies

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A R T I C L E I N F O

Keywords:
Anti-infective agents
Anti-bacterial agents
Chronic disease
Gene expression
Virulence factors

A B S T R A C T

Microbial biofilms serve an important function in various aspects of human health, including infections, chronic diseases, and industrial processes, but their development poses significant challenges due to inherent resistance to traditional antimicrobial agents. Natural compounds are increasingly being investigated as potential anti-biofilm therapy. Resveratrol, a polyphenolic molecule found in many plants, has sparked widespread interest since it has numerous physiological properties, namely antibacterial characteristics. The purpose of this paper is to conduct a thorough analysis of the effect of resveratrol on microbial biofilm. We will elaborate on how resveratrol reduces biofilm formation, disrupts pre-formed biofilms, and modulates gene expression related to biofilm development. Furthermore, we discuss the effects of resveratrol on the structural integrity, composition, and virulence factors associated with biofilms. Additionally, we explore the potential synergistic interactions between resveratrol and other antimicrobial agents, while highlighting the challenges and future prospects of utilizing resveratrol as an effective anti-biofilm agent. Overall, the present investigation offers details on promising benefits of resveratrol in eliminating microbial biofilms, emphasizing its efficacy as an alternative treatment for biofilm-associated illnesses and their consequences.

1. Introduction

Microbial biofilms are increasingly recognized as a major contributor to persistent infections and antibiotic tolerance across diverse anatomical sites [1]. These structured microbial communities pose significant challenges to clinical management due to their resilience and adaptability [2]. This review aims to synthesize current knowledge on anti-biofilm mechanisms of resveratrol, with a particular focus on its pharmacological diversity and integration into advanced drug delivery systems. The review opens with a summary of biofilm biology and its clinical significance, followed by a system-specific examination of biofilm-related infections and the anti-biofilm potential of resveratrol. Bridging molecular mechanisms with translational strategies, the manuscript outlines a comprehensive framework for future therapeutic developments targeting resilient microbial communities in biomedical contexts.

1.1. Microbial biofilms and its clinical implications

Microbes can exist as isolated individuals or as members of a community known as biofilms. Biofilms are organized populations of microorganism that exist within an extracellular polymeric substance (EPS) matrix. In contrast to their freely floating planktonic counterparts, these biofilms show different growth patterns and genetic expression profiles and attach to both living and non-living surfaces [3–5]. Biofilms enable microorganisms to form symbiotic connections with hosts, withstand life-threatening environmental circumstances, and develop resistance to antibiotics and other external stimuli [6]. Biofilm formation plays a significant role in the advancement of antibiotic resistance and persisting cell populations, that contributes to the challenging

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https://doi.org/10.1016/j.fitote.2025.106800
Received 21 May 2025; Received in revised form 21 July 2025; Accepted 7 August 2025
Available online 8 August 2025

persistence of microbial infections [7].

Biofilms exhibit an extensive variety of clinical symptoms and can be found in a number of habitats, including food processing units, hospital facilities, living tissues, medical implants, pipes, water channels, as well as various surfaces that are biotic or abiotic in nature [8,9]. Microorganisms in biofilms modify their phenotypes and gene expression, resulting in resistance to conventional antibiotics, declined growth rates and metabolic activity, and the generation of virulence-associated proteins [3,10]. Approximately 80 % of chronic illnesses and 65 % of microbiological infections are caused by microbial biofilms, according to research from the National Institutes of Health.

These biofilms can affect both tissues and various medically implanted devices. Examples of such devices prone to infection by microbial biofilms include breast implants, catheters (such as endotracheal and urinary catheters), contact lenses, defibrillators, joint prostheses, mechanical heart valves, orthopedic implants, pacemakers, tissue fillers, vascular grafts, ventricular-assisted devices, ventricular shunts, and voice prostheses [4].

Importantly, biofilm formation contributes to a wide spectrum of tissue-specific infections. In the oral cavity, biofilms are central to the pathogenesis of dental plaque, periodontitis, and chronic tonsillitis [11,12]. In ocular tissues, biofilm-related infections include bacterial keratitis and conjunctivitis, often associated with contact lens use [13]. Respiratory biofilms are implicated in chronic rhinosinusitis and cystic fibrosis-associated lung infections [14], while digestive system biofilms contribute to *Helicobacter pylori*-related gastritis and colorectal infections [15]. In the genitourinary tract, biofilms underlie recurrent urinary tract infections and catheter-associated complications [16]. Cutaneous biofilms are involved in chronic wound infections and acne vulgaris [17]. These infections are often refractory to conventional antibiotics due to the protective biofilm matrix and altered microbial physiology [18].

The anatomical and physiological differences among organ systems influence the pathogenic mechanisms of biofilm formation and the nature of associated infections [1]. These distinctions underscore the need for tailored therapeutic approaches, including natural compounds like resveratrol, which may exert system-specific anti-biofilm effects.

Moreover, biofilm-driven antibiotic tolerance mechanisms—such as reduced drug penetration, efflux pump activation, and metabolic dormancy—further complicate treatment [18]. Addressing these mechanisms is essential for developing effective anti-biofilm agents. These substances should aim at particular signaling pathways that are important for the formation of biofilms, such as adhesion, dispersion, EPS generation, genes linked to biofilms, quorum sensing, microbial motility, and other key mechanisms [19–21]. Disrupting the production of biofilm and improving the effectiveness of alternative therapies against disorders linked to biofilms can be achieved by focusing on these essential processes. In order to successfully address the issues posed by these tough microbial communities, it is very promising to develop innovative anti-biofilm agents.

1.2. Natural compounds as anti-microbial agents

Given the well-known biological and pharmacological activities of medicinal plants and phytochemicals [22–25], numerous studies have focused on the impact of these natural products on microbial species and associated infections. The results of previous investigations have demonstrated that some herbs and their active constituents such as *Zingiber officinale* [26], *Thymus vulgaris* [27], *Syzygium aromaticum* [28], *Origanum vulgare* [29], *Rosmarinus officinalis* [30], *Allium sativum* [31], rosmarinic acid [32], and berberine [33] are effective anti-microbial agents.

1.3. Resveratrol and its pharmacological effects

Resveratrol was the first stilbenoid compound discovered in *Veratrum grandiflorum*, commonly known as white hellebore [34,35]. Like other stilbenes, resveratrol is produced in response to a variety of stimuli, including direct ozone exposure, pathogen response, and ultraviolet light exposure [36,37]. This compound is naturally found in different plant species including bananas, beans, grapes, peanuts, pine trees, pomegranates, and soybeans [38–40]. Numerous pharmacological activities have been reported for resveratrol [41–43]. These include immune system regulation, inflammation reduction, cancer prevention, brain and heart protection, and the treatment of diseases like cancer, Parkinson's disease, and diabetes [41,44–47] though conflicting data particularly from clinical studies also exist [48–52]. Resveratrol has also shown antiviral, antifungal, and antibacterial properties [53,54]. Moreover, recent advancements in therapeutic approaches have highlighted the photosensitive and sonosensitive properties of resveratrol, unveiling its potential applications in photodynamic and sonodynamic therapy [55–57]. In addition to its well-documented effects on human health, emerging evidence suggests that resveratrol may possess antimicrobial properties, making it a promising candidate for opposing microbial biofilms.

In this review, we aim to comprehensively assess the impact of resveratrol on microbial biofilms. We will explore the mechanisms of action by which resveratrol inhibits biofilm formation, disrupts preformed biofilms, and modulates biofilm-related gene expression. We will also examine how resveratrol affects biofilm composition and structural integrity, as well as how it affects virulence variables linked to biofilms. Additionally, we will discuss the potential synergistic interactions of resveratrol with other antimicrobial agents and highlight the challenges and future prospects of utilizing resveratrol as an antibiofilm agent.

Understanding the potential of resveratrol in eliminating microbial biofilms is critical because it may open the way for discovering innovative treatment techniques to attenuate infections caused by biofilm and related consequences. This study seeks to give useful insights into the processes and effects of resveratrol on microbial biofilms by explaining its potential as an effective and sustainable method to managing biofilm-related challenges.

2. Methods

For this review, multiple search engines and databases, for example PubMed, Science Direct, Scopus, and Google Scholar, were used. The search encompassed articles published up until October 2024, without any time constraints. The search terms included "resveratrol", "microbial biofilms", "antibiofilm", "antimicrobial", "biofilm inhibition", "biofilm formation", "Quorum sensing", "biofilm dispersal", "biofilm matrix", "human body systems", and "microbial adhesion".

3. The impact resveratrol on human biofilm formation and control

3.1. Effect of resveratrol on oral cavity and dental structures

Microbial biofilms have a significant role in oral and dental health, and their effect extends to dental implants [58]. Biofilms occur on numerous surfaces within the oral cavity, including natural teeth, dental restorations, and implants [59]. A self-produced extracellular matrix protects many microbial communities that make up these biofilms. The sticky substance known as dental plaque, which sticks to tooth surfaces, is formed in part by biofilms [60]. Dental plaque biofilms can cause a variety of oral health problems, such as dental caries (tooth decay) and periodontal disorders including gingivitis and periodontitis [61]. Dental implant biofilms may potentially be an issue. Peri-implant illnesses, including as peri-implant mucositis and peri-implantitis, may result from them. Inflammation of the soft tissues around the implant is known as peri-implant mucositis, while bone loss is known as peri-implantitis. These biofilms have the potential to affect the success and lifespan of

Table 1
Effect of resveratrol on microbial biofilm and its implications for oral cavity, dental, and ocular structures.

Disorder/Condition	Study design	Doses	Results	Ref.
Oral and dental health				
In vitro				
Periodontal disease	Oral epithelial cell, *A. actinomycetemcomitans*, *F. nucleatum*, *P. gingivalis*, and *S. mitis*	0.01, 0.05, 0.5 % *w/v*	-No bactericidal effect ↓ Releasing inflammatory mediators from epithelial cells	[63] 2014
Dental plaque biofilm formation	*F. nucleatum*	0, 12.5, 25, 50, and 100 μg/ml	-No significant effect on bacterial growth rate ↓Biofilm formation, gene expression of the biofilm	[64] 2016
Periodontal disease	*P. gingivalis*	78.12–156.25 μg/ml	-Inhibited the biofilm formation ↓ virulence of *P. gingivalis*, gene expression of and proteinases (kgp and rgpA), fimbriae (type II and IV)	[65] 2019
Porphyromonas gingivalis	*P. gingivalis* ATCC 33277	15.63–500 μg/ml	- Destroyed pre-formed *P. gingivalis* biofilm ↑ Damage to the bacterial cell membrane, activation of NF-κB signaling pathway ↓ Bacterial adherence to matrix proteins, breakdown of keratinocyte tight junctions, protease activities of *P. gingivalis*, soluble TREM-1 secretion in monocytes, TREM-1 gene expression	[66] 2019
Dentin bonding durability	Human third molars	10 mg/ml	↑ Microtensile bond strength ↓ Expression of nanoleakage, matrix metalloproteinases and *S. mutans* activity, cohesive failure in dentin	[67] 2020
Implant-associated infections	*P. aeruginosa* PAO1, and *S. mutans*	200 μM	↑ Antibacterial and antibiofilm activity against *P. aeruginosa* PAO1, and *S. mutans*	[68] 2020
Dental caries	*S. mutans*	0, 50, 100, 200, 400 μg/ml	↓ Acid tolerance and production, biofilm formation, virulence gene expression (ldh, relA, gtfC, comDE)	[69] 2020
Periodontitis	*F. nucleatum*	1.25 mg/ml	↑ Leakage of cellular contents, antioxidant pathways ↓ Bacterial cell viability, NF-κB activation	[70] 2020
Composite restorations	*S. mutans*	1 mg/ml	- Preserved bond strength ↓ Nanoleakage expression, *S. mutans* biofilm formation, endogenous protease activity	[71] 2021
Denture stomatitis	Human gingival fibroblast	512 μg/ml	↓ Gene expression of TNF-α and IL-6, microbial biofilm growth	[73] 2023
In vivo				
Periodontitis	Male Wistar rats	10 mg/kg, 30 days, gavage	-No significant effect on the levels of IL-1β and IL-4 ↓Bone-loss values, IL-17	[74] 2013
Periodontitis	Male Wistar rats	10 mg/kg, 30 days, gavage	- No significant effect on the levels of *A. actinomycetemcomitans, P. gingivalis*, and *T. forsythia*	[75] 2016
Periodontitis	C57BL/6 N wild-type mice	10 mg/kg, i.p.	↑ Periodontal bone healing, Nrf2 ↓ IL-1β	[76] 2022
Clinical trial				
Gingival inflammation in early childhood	64 children	Once daily, after tooth brushing, 4 weeks	↑ Salivary pH ↓ Percentage of bleeding sites, dental plaque	[77] 2021
Chronic periodontitis	40 patients	480 mg, 4 weeks	↓ Plaque index	[78]
Periodontitis	57 individuals	10 ml, twice a day, 30 days	↓ Bleeding on probing, clinical attachment loss, plaque index, probing pocket depth, salivary IL-6 and RANKL levels	[79]
Ocular health				
In vitro				
Biofilm formation on contact lenses	*P. aeruginosa* and *S. aureus*	100 μg/ml in ethanol:water 10:90 *v*/v	↑ Antimicrobial activity against *P. aeruginosa* and *S. aureus* ↓ Biofilm formation	[83] 2021
Biofilm formation on contact lenses	*P. aeruginosa* and *S. aureus*	100 μg/ml in ethanol:water 10:90 v/v	↓ Growth of *P. aeruginosa* and *S. aureus*	[84] 2022
Ocular health	*P. aeruginosa* and *S. aureus*	10, 50, 100, 200 and 300 μg/ml	↑ Antimicrobial activity against *P. aeruginosa* and *S. aureus*	[85] 2022

IL: interleukin; NF-κB: nuclear factor kappa-light-chain-enhancer of activated B cells; Nrf2: nuclear factor erythroid 2-related factor 2; RANKL: receptor activator of nuclear factor-kappa B ligand; TNF-α: tumor necrosis factor-alpha; TREM-1: triggering receptor expressed on myeloid cells-1.

dental implants [62]. Understanding the significance of microbial biofilms in oral and dental health, as well as their influence on dental implants, is critical in developing effective preventative and management strategies for maintaining oral health and implant longevity.

3.1.1. In vitro studies on oral cavity and dental structures

An investigation was conducted to explore the inflammatory response of epithelial cells towards oral biofilms (*Aggregatibacter actinomycetemcomitans*, *Fusobacterium nucleatum*, *Porphyromonas gingivalis*, and *Streptococcus mitis*) when cultured together, as well as the potential role of resveratrol in moderating this response. The application of resveratrol to epithelial cells was found to effectively alleviate inflammation at the protein level. However, it had no bactericidal effect [63]. Treating *F. nucleatum*, anaerobic bacterium that has a chief role in the formation of dental plaque, with resveratrol decreased biofilm formation and gene expression of the biofilm. However, it had no significant effect on bacterial growth rate [64]. The effect of resveratrol was assessed on *P. gingivalis* in periodontal disease. The results showed that

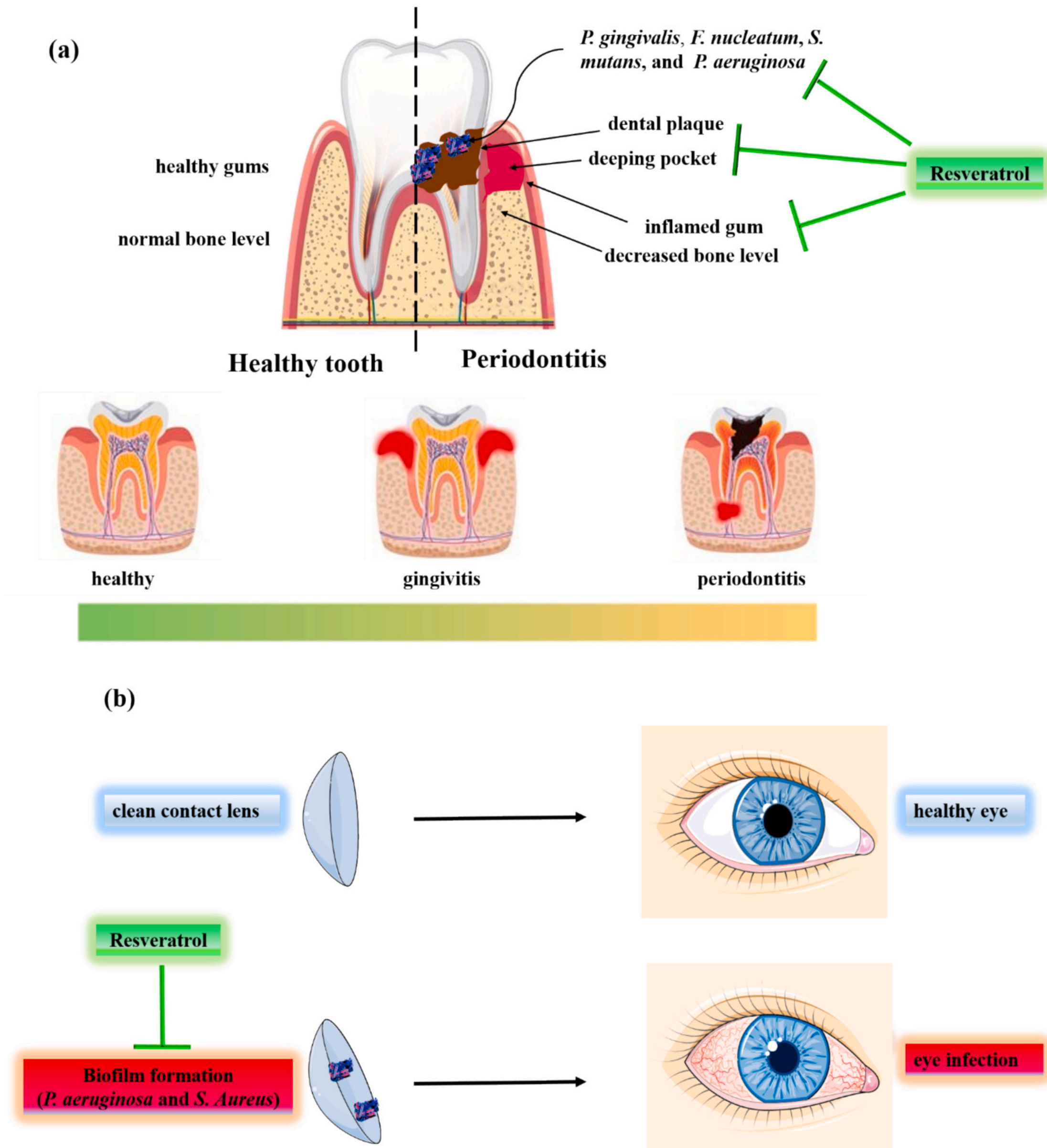

Fig. 1. (a) The ameliorative effect of resveratrol on periodontitis; (b) The therapeutic effect of resveratrol on eye infection (Images from www.freepik.com and https://smart.servier.com).

resveratrol reduced the virulence and prevented the production of biofilms in *P. gingivalis*. It also decreased the expression of the genes for the proteinases rgpA and kgp, as well as the fimbriae type II and IV [65]. Furthermore, resveratrol exhibited antibacterial effects, which might be attributable to its ability to destroy the bacterial cell membrane. Additionally, it was found to decrease bacterial attachment to matrix proteins and effectively eliminate pre-formed *P. gingivalis* biofilm. Also, resveratrol decreased the degradation caused by *P. gingivalis*, protecting the integrity of keratinocyte tight connections. The ability of resveratrol to inhibit *P. gingivalis* protease activity may be the cause of this protective effect. Furthermore, resveratrol decreased the expression of the gene for triggering receptor expressed on myeloid cells-1 (TREM-1) and soluble TREM-1 release in monocytes, as well as the activation of the nuclear factor kappa B (NF-B) signaling pathway triggered by *P. gingivalis* [66]. Pretreatment of sound human third molars with resveratrol/ethanol solutions increased microtensile bond strength, and reduced the expression of nanoleakage, matrix metalloproteinases and *Streptococcus mutans* activity, as well as cohesive failure in dentin [67]. Moreover, it has been disclosed that polylactic acid-resveratrol2 (a specific formulation of polylactic acid containing 3.2 % weight of resveratrol) exhibited noteworthy antibiofilm and antibacterial properties against *S. mutans* and *Pseudomonas aeruginosa* PAO1, the strains cause both acute and chronic infections in humans including implant-associated infections [68]. Resveratrol had notable effects on the activity of *S. mutans*. It inhibited the synthesis of both water-soluble and water-insoluble polysaccharides, prevented the formation of biofilms, and dramatically decreased acid production and acid tolerance. Additionally, resveratrol inhibited the expression of virulence genes such as comDE, gtfC, ldh, and relA [69]. Another *in vitro* study compared the antimicrobial properties of resveratrol and its analogues, namely oxyresveratrol, piceatannol, and pterostilbene. With a minimum inhibitory concentration (MIC) that is sixty times smaller than that of resveratrol, oxyresveratrol, and piceatannol, the results demonstrated that pterostilbene had the strongest antibacterial effects. Furthermore, pterostilbene effectively inhibited the formation of *F. nucleatum* biofilms. The antimicrobial action of pterostilbene was attributed to the release of cellular contents, resulting in the loss of bacterial viability. Moreover, pterostilbene demonstrated immunomodulatory effects on *F. nucleatum*-challenged macrophages by upregulating antioxidant pathways and inhibiting nuclear factor kappa-light-chain-enhancer of activated B cells (NF-κB) activation [70]. The resveratrol-doped adhesive effect was assessed on the durability and antibiofilm capability of dentin bonding. The findings illustrated that regardless of whether the resveratrol-doped adhesive underwent collagenase ageing or thermocycling, it consistently minimized nanoleakage expression and conserved bond strength. Furthermore, while preserving acceptable biocompatibility, the adhesive showed the capacity to suppress endogenous protease activity and the development of *S. mutans* biofilms [71]. Another study was carried out to determine the proteins in red complex organisms that bind to resveratrol. The proteins that interacted with resveratrol were primarily involved in cellular processes, followed by metabolism and virulence factors. Notably, no proteins associated with information storage were detected. Interestingly, according to VirulentPred scores, carboxynorspermidine decarboxylase and Superoxide dismutase Fe—Mn were identified as virulence factors. Additionally, two compounds, namely pyridine nucleotide-disulfide oxidoreductase and a hypothetical protein, were found to be virulent based on their scores obtained from VirulentPred, associated with metabolism and cellular processes, respectively. The majority of proteins that interacted with resveratrol were categorized under cellular processes, followed by metabolism and virulence factors. One protein, serpin, associated with metabolism, and a protein, carboxynorspermidine decarboxylase, were predicted to be linked to virulence [72]. It has been reported that treating human gingival fibroblast infected with *Actinomyces naeslundii*, *Candida albicans*, *Staphylococcus aureus*, and *Streptococcus sobrinus* with nano-resveratrol could remarkably reduce the gene expression of interleukin (IL)-6 and tumor necrosis factor-alpha (TNF-α), as well as microbial biofilm growth [73].

3.1.2. In vivo studies on oral cavity and dental structures

It has been reported that administration of resveratrol might ameliorate periodontal breakdown in rats by reducing bone-loss values and the levels of IL-17. But it had no significant effect on the levels of IL-1β and IL-4 [74]. The administration of resveratrol to rats with periodontitis disclosed no significant effect on the levels of *Tannerella forsythia*, *A. actinomycetemcomitans,* and *Porphyromonas gingivalis* [75]. The administration of resveratrol, whether in dimer or monomeric form, was found to effectively decrease periodontal bone loss. The dimer group exhibited greater inhibition of bone loss compared to the monomer group. These beneficial effects were likely attributed to a reduction in oxidative stress (increasing the amount of nuclear factor erythroid 2-related factor 2 (NRF2)), leading to a decrease in local inflammation (reducing the amounts of IL-1β) [76].

3.1.3. Clinical trials on oral cavity and dental structures

Using resveratrol-nanovector of 2-hydroxypropyl-beta-cyclodextrins spray increased salivary pH, and reduced the percentage of bleeding sites as well as dental plaque in children with gingival inflammation and bacterial biofilm [77] (Table 1). Another investigation assessed how resveratrol supplementation affected inflammatory markers and clinical characteristics in patients with chronic periodontitis. Over the course of four weeks, the individuals received non-surgical treatment and were randomized to receive either resveratrol supplements or a placebo. Salivary levels of IL-8 and IL-1β were measured, along with important clinical markers such as bleeding index, clinical attachment level, plaque index, and pocket depth. Only the plaque index showed a significant difference between the resveratrol and control groups after the intervention, according to the results; other metrics and salivary markers did not [78]. The effectiveness of resveratrol mouthwash as a supplementary to nonsurgical periodontal therapy for periodontitis was evaluated in a double-blind clinical trial. The results demonstrated that resveratrol and chlorhexidine significantly improved bleeding on probing, clinical attachment loss, plaque index, and probing pocket depth surpassing a placebo. Resveratrol and chlorhexidine mouthwashes did not significantly differ in their ability to lower salivary IL-6 levels. Receptor activator of nuclear factor-kappa B ligand (RANKL) levels dropped equally among all groups [79].

Resveratrol studies show that it has promising oral health advantages. Resveratrol lowers inflammation and inhibits biofilm-related components in oral bacteria such as *P. gingivalis* and *F. nucleatum*. It destroys bacterial membranes, inhibits virulence genes, and opposes infections such as *S. mutans*, which helps to prevent plaque formation and caries. Resveratrol improves periodontal health by decreasing bone loss and inflammation (Fig. 1a). Clinical trials with resveratrol-nanovectors reveal improved salivary pH, fewer bleeding sites, and less plaque accumulation, indicating potential for treating oral problems. More research is needed to determine the optimal dosage, administration, and long-term benefits of resveratrol on dental health.

3.2. Effect of resveratrol on ocular structures

Microbial biofilms have an important role in ocular health, particularly when contact lenses or ocular lenses are used. Because contact lenses are in prolonged contact with the ocular surface, they provide an optimal environment for the production of microbial biofilms. The presence of bacteria, fungus, or other microorganisms on lenses can form biofilms, which can cause a number of ocular complications [80]. These complications include contact lens-related microbial keratitis, inflammation of the cornea, and other lens-related infections [81]. Microorganisms can be shielded from antimicrobial agents by the biofilm matrix, which increases their resistance to destruction [82]. Understanding the role of microbial biofilms on ocular surfaces is therefore critical for developing effective strategies for preventing and managing lens-related infections, maintaining ocular health, and promoting safe contact lens usage.

3.2.1. In vitro studies on ocular structures

Resveratrol-loaded hydrogel contact lenses showed antimicrobial activity against *S. aureus* and *P. aeruginosa*. They also reduced the formation of biofilm [83]. Likewise, phosphorylcholine-based contact lenses for sustained release of resveratrol were designed and the findings revealed that they could remarkably reduce the growth of *P. aeruginosa* and *S. aureus* [84] (Fig. 1b). Resveratrol nanoparticles (single and mixed micelles) exhibited antimicrobial effects against *S. aureus* and *P. aeruginosa*. In the *ex vivo* part of the study, these micelles demonstrated a preference for accumulating resveratrol in the cornea and sclera. However, compared to single micelles, mixed micelles exhibited longer delay periods and resulted in reduced permeation of resveratrol through the sclera [85] (Table 1).

Resveratrol-based developments, such as hydrogel contact lenses, phosphorylcholine-based lenses, and nanoparticles, have antibacterial activity against *P. aeruginosa* and *S. aureus*. These methods limit biofilm formation and bacterial growth. Resveratrol nanoparticles, particularly mixed micelles, aggregate effectively in the cornea and sclera, indicating the possibility for targeted drug administration. Further study is needed to optimize formulations and investigate long-term effects, but these findings provide significant opportunities for enhanced antibacterial treatments in ocular care.

3.3. Effect of resveratrol on respiratory system

Microbial biofilms are important for respiratory system health. Biofilms can form on a variety of surfaces in the respiratory system, including the airway epithelium, trachea, bronchi, and lung tissues [86]. These biofilms, which can include bacteria, fungus, and viruses, have been linked to chronic respiratory diseases such as cystic fibrosis, chronic obstructive pulmonary disease (COPD), and ventilator-associated pneumonia [87]. Biofilms contribute to the development of

Table 2
Effect of resveratrol on microbial biofilm and its implications on respiratory system.

Disorder/ Condition	Study design	Doses	Results	Reference
In vitro				
Cystic fibrosis	*P. aeruginosa* PA14	10 μg/ml	↓Biofilm formation, pyocyanin production	[93] 2013
Lung infection	*P. aeruginosa* and *C. violaceum*	25, 50, 100, and 200 μg/ml	↑ Susceptibility to antibiotics ↓ Quorum sensing-mediated violacein pigment production in *C. violaceum*, biofilm formation, proteolytic activity, pyocyanin production, and swarming motility in *P. aeruginosa*	[94] 2017
Cystic fibrosis	*P. aeruginosa*	50, 100, and 150 μg/ml	↑ Susceptibility to aminoglycoside antibiotics, biofilms architectural disruption ↓ lasI and rhlI gene expression	[95] 2018
Lung infection	BEAS-2B cells, *P. aeruginosa*	20 μM	-Inhibited the reduction of TFAM expression by quorum sensing molecules, the early (3−12h) drop in TEER triggered by 3-oxo-C12-HSL - Conserved tight junction architecture, barrier integrity ↓ Transmigration of bacteria across the epithelial barrier, mitochondrial biogenesis impairment	[96] 2019
Nasopharyngeal cavity infection (Pneumococcal infections)	*S. pneumoniae*	10 to 50 μM	↓ pre-established *in vitro* biofilms and number of bacteria, membrane integrity	[97] 2019
Lung infection	*K. pneumoniae*	32 to 128 μg/ml	↑ Killing bacteria ↓ Minimum inhibitory concentrations against polymyxin B	[98] 2020
Lung infection	*P. aeruginosa*	5–20 ng/ml	↑ Antimicrobial effect of Tobramycin	[99] 2021
Lung infection	*P. aeruginosa*		↑ Anti-biofilm and antibacterial activities	[100] 2022
Cystic fibrosis	Airway organoids from lung of a cystic fibrosis patient, *M. abscessus*	10 μM	↑ Control of acteria growth ↓S and R variant growth in airway organoids	[101] 2022
Cystic fibrosis	Airway organoids from lung of a cystic fibrosis patient, *M. abscessus*	10 μM	↓S and R variant growth in airway organoids	[102] 2023

TEER: transepithelial electrical resistance; TFAM: mitochondrial transcription factor-A.

these diseases through enabling the exchange of genetic material and antimicrobial resistance genes among microorganisms [88,89]. Moreover, biofilms can cause structural damage to the respiratory tissues, impair mucociliary clearance, and trigger exaggerated immune responses, leading to chronic inflammation and tissue damage [90–92]. Determining the significance of microbial biofilms in respiratory system health is critical for developing effective ways for preventing and treating respiratory infections as well as improving overall lung function.

3.3.1. *In vitro studies on respiratory system*

It has been demonstrated that ε-viniferin inhibited biofilm formation and reduced pyocyanin production in *P. aeruginosa* PA14 [93]. Research used the *Chromobacterium violaceum* biosensor bioassay to detect the anti-quorum sensing activity of a new resveratrol formulation called Resveramax™ (1 % grape seed oil, 1 % sunflower lecithin, and 50 % resveratrol). Standard methods were then used to evaluate the impact of Resveramax™ on quorum sensing-regulated phenotypes in *P. aeruginosa* PAO1. The generation of violacein pigment in *C. violaceum*, biofilm formation, proteolytic activity, pyocyanin production, and swarming motility in *P. aeruginosa* PAO1 were all specifically and concentration-dependently suppressed by Resveramax™. When biofilms were treated with Resveramax™ instead of only antibiotics, the biofilms became more susceptible to the antibiotics [94]. Researchers investigated how different antibiotics combined with the quorum sensing inhibitor resveratrol affected *P. aeruginosa* PAO1 biofilms. The biofilms of *P. aeruginosa* PAO1 that were grown with resveratrol were found to be more susceptible to aminoglycoside antibiotics. The combined treatment of resveratrol and aminoglycoside antibiotics resulted in significant architectural disruption of the biofilms. Additionally, it was revealed that the expression of lasI and rhlI genes, responsible for the synthesis of signal molecules in quorum sensing systems, was effectively inhibited in *P. aeruginosa* PAO1 biofilms by resveratrol [95]. Treating BEAS-2B cells infected with *P. aeruginosa* resulted in conserved tight junction architecture and barrier integrity. Resveratrol also reduced the transmigration of bacteria across the epithelial barrier and mitochondrial biogenesis impairment [96]. It has been reported that viniferin (resveratrol oligomers) revealed its antibiofilm and antimicrobial properties against *Streptococcus pneumoniae* by reducing pre-established *in vitro* biofilms, number of bacteria, and membrane integrity [97]. Furthermore, the findings of another investigation disclosed that resveratrol augmented the antimicrobial property of polymyxin B on *Klebsiella pneumoniae* by increasing polymyxin B ability to kill bacteria and decreasing the minimum inhibitory concentrations against polymyxin B [98]. Another research demonstrated the efficacy of using Eudragit nanoparticles comprising E100/P188 and NE30D/P407 to transport Tobramycin and resveratrol, respectively. By combining these components in equal proportions, an innovative binary fluid was generated, augmenting the antimicrobial effectiveness of antibiotic. This was evident in the minimum inhibitory concentrations and swarming inhibition tests conducted against *P. aeruginosa* [99]. It has also been shown that *co*-administration of resveratrol and polymyxin B exhibited synergistic anti-biofilm and antibacterial properties against *P. aeruginosa* [100]. Treating airway organoids from lung of a cystic fibrosis patient infected with *Mycobacterium abscessus* increased the

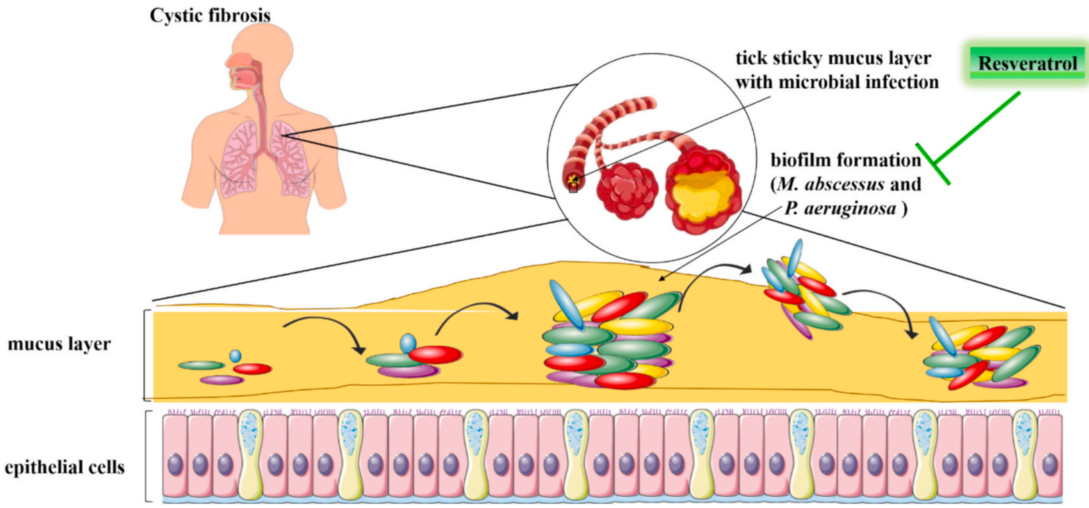

Fig. 2. The healing effect of resveratrol on cystic fibrosis (Images from www.freepik.com and https://smart.servier.com).

control of bacteria growth and attenuated S and R variant growth in airway organoids [101,102] (Table 2) (Fig. 2).

In summary, research demonstrates the potential of resveratrol against a variety of bacterial infections. Resveratrol and its formulations prevent biofilm formation and infections caused by *P. aeruginosa*, *S. pneumoniae*, *K. pneumoniae*, and *M. abscessus*. They inhibit *P. aeruginosa* quorum sensing, increase antibiotic efficacy, break biofilm formations, and strengthen epithelial cell defenses. Nanoparticles containing tobramycin and resveratrol enhance antibacterial activity. Combinations of aminoglycosides and polymyxin B have synergistic

Table 3
Effect of resveratrol on microbial biofilm and its implications on digestive system.

Disorder/Condition	Study design	Doses	Results	Reference
In vitro				
Colonic epithelial cell infection	Colonic epithelial cell lines, *E. coli*, *L. monocytogenes*, and *S. typhimurium*	25–100 μM	↓ Adhesion of *E. coli* and *S. typhimurium*, secretion of IL-8	[107] 2012
Gastrointestinal infection	*E. coli* and L. *monocytogenes*	60, 80, 100, and 120 μg/ml	-No significant antimicrobial effect ↓Violacein production	[108] 2012
Gastrointestinal infection	*E. coli*	10 μg/ml	-No significant effect on biofilm formation in four commensal *E. coli* strains	[109] 2013
Cholera	*V. cholerae*	10–30 μg/ml	↓ Biofilm formation	[110] 2014
Bacterial gastroenteritis	*A. butzleri*, *C. coli*, and *C. jejuni*	64–256 μg/ml	-Demonstrated anti-Arcobacter and anti-Campylobacter properties, anti-quorum sensing activity ↑ Biofilm dispersion ↓ Biofilm formation	[111] 2015
Gastrointestinal infection	*A. butzleri*, *C. coli*, and *C. jejuni*	0.25 % and 75 % *w/v*	-Showed anti-quorum sensing activity ↓ Biofilm formation	[112] 2016
Gastroenteritis	*L. monocytogenes*	200 μg/ml	↓ Biofilm formation	[116] 2016
Gastrointestinal infection	*C. albicans*, *E. coli*, *P. aeruginosa*, and *S. epidermidis*	100, 160, 170 mg/l	↑Disrupt preformed biofilms ↓ Biofilm formation	[117] 2018
Peptic ulcer	*H. pylori*		-Synergize with levofloxacin ↑ Antibacterial activity ↓ Biofilm formation, swarming motility	[118] 2020
Enhancing probiotics function	*L. paracasei*	10–50 μM	- Adjusted physico-chemical properties of the bacterial surface ↑ Adhesion and biofilm formation	[119] 2020
Gut infection	*L. fermentum*, *S. enterica*	31.25–2000 μg/ml	↑Probiotic properties of L. *fermentum* ↓Growth of *S. enterica*	[120] 2021
Intestinal infection	*E. coli*		↓ Growth of *E. coli*	[121] 2022
In vivo				
Enterocolitis	IL-10^{-}/$^{-}$ mice	60 mg/kg, 4 days, p. o.	↑Ameliorated clinical conditions, function of colonic epithelial barrier ↓Colonic epithelial apoptosis, NO secretion	[122] 2020
Liver fibrosis	Balb/c mice	30 mg/kg, 4 weeks, gavage	↓ Growth of *S. lentusand* and *S. xylosus*	[123] 2022

IL: interleukin; NO: nitric oxide.

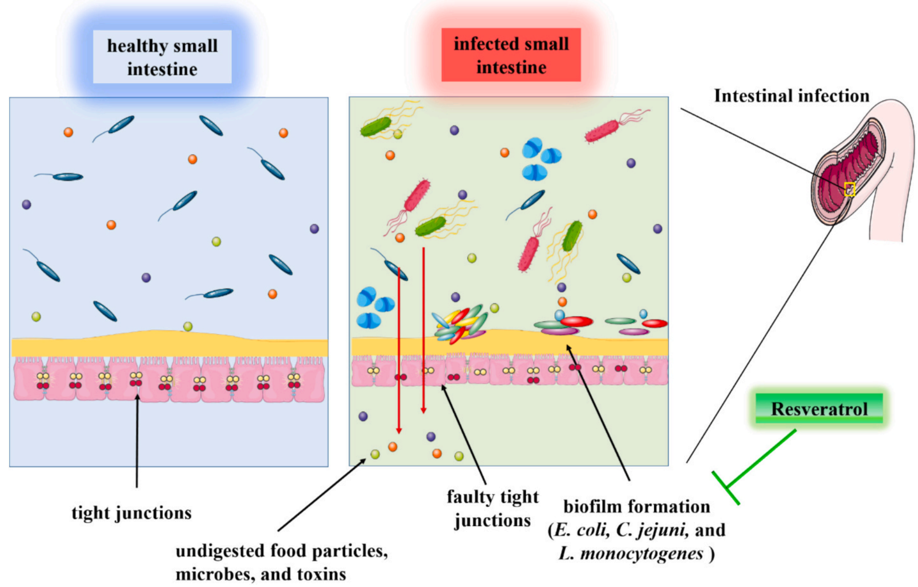

Fig. 3. The protective effect of resveratrol on small intestine microbial biofilm (Images from www.freepik.com and https://smart.servier.com).

effects against *P. aeruginosa*. Resveratrol therapy inhibits *M. abscessus* development in airway organoids, suggesting new antibacterial treatments for bacterial infections.

3.4. Effect of resveratrol on digestive system

Biofilms affect several parts of the digestive system and are important for gastrointestinal health. These complicated networks of bacteria establish a protective matrix by adhering to the mucosal surfaces of the gastrointestinal system. The presence of biofilms can affect gut health in both favorable and negative ways. Positively, biofilms support the growth of a healthy gut microbiota by providing beneficial bacteria a stable habitat in which to thrive. They facilitate the digestion of complex carbohydrates, enhance nutrient absorption, and generate advantageous metabolites such short-chain fatty acids. Furthermore, biofilms support the intestinal barrier and keep harmful organisms from penetrating the gut lining [103]. On the other hand, biofilms might have harmful effects. They can host harmful strains of bacteria and contribute to chronic infections or antibiotic resistance. [104]. In addition, biofilms have the potential to disrupt the delicate balance of the gut microbiota, which can result in dysbiosis and related gastrointestinal disorders [105,106]. In order to prevent or treat gastrointestinal disorders and maintain a balanced gut microbiota, it is important recognize the role of biofilms in gastrointestinal health.

3.4.1. In vitro studies on digestive system

An *in vitro* study was conducted to examine the effectiveness of resveratrol and various derivatives (glucuronide, glucosylacyl, and glucosyl) in preventing the attachment of *Salmonella typhimurium*, *Escherichia coli*, and *Listeria monocytogenes* Scott A to Caco-2 and HT-29 colonic cells. The application of resveratrol and most of its derivatives resulted in a decrease in adhesion of *E. coli* and *S. typhimurium* to colonic cells. However, the inhibition of adhesion was less pronounced when dealing with L. *monocytogenes*, a bacterium known for its higher adherence potential. Specifically, resveratrol-3-O-(6′-Obutanoyl)-β-D-glucopyranoside and resveratrol-3-O-(6′-O-octanoyl)-β-D-glucopyranoside demonstrated a complete reduction in IL-8 secretion [107]. Treating *E. coli* and L. *monocytogenes* with resveratrol decreased violacein production, but it had no significant antimicrobial effect on the tested concentrations [108]. Likewise, treating *E. coli* with trans-resveratrol had no significant effect on biofilm formation in four commensal *E. coli* strains [109]. The results of another study demonstrated that resveratrol at subinhibitory concentrations effectively suppressed the formation of biofilms in *Vibrio cholerae*. Docking analysis identified AphB as the potential target of resveratrol [110]. Other investigations assessed the effect of resveratrol hydroxypropyl-γ-cyclodextrin inclusion complexes on *Arcobacter butzleri*, *Campylobacter coli*, and *Campylobacter jejuni*. The obtained data indicated that this compound demonstrated anti-Arcobacter and anti-Campylobacter properties, anti-quorum sensing activity, increased biofilm dispersion, and reduced biofilm formation [111,112]. Resveratrol also exhibited antimicrobial and anti-biofilm properties against *Listeria monocytogenes*. The antimicrobial properties of resveratrol were observed in both planktonic cells and their ability to form biofilms. Resveratrol has been proposed to possess various potential applications in the food industry [113–115]. To explore its potential as a food preservative, the effectiveness of resveratrol was assessed in various food models, including milk, lettuce leaf, and chicken juice. Resveratrol demonstrated higher efficacy in the lettuce leaf and chicken juice models, while its antilisterial activity was negatively influenced by milk, suggesting a potential decrease in resveratrol availability in milk [116]. Furthermore, resveratrol disrupted performed biofilms and decreased biofilm formation of *C. albicans*, *E. coli, P. aeruginosa,* and *Staphylococcus epidermidis* [117] (Table 3). An experiment was conducted to evaluate the antivirulence and antibacterial characteristics of resveratrol and its novel derivatives. The study also examined their ability to enhance the antibacterial effects of levofloxacin. Among the different resveratrol derivatives analyzed, resveratrol-3 and resveratrol-4 exhibited stronger antibacterial activity compared to resveratrol. Additionally, it was discovered that levofloxacin and resveratrol, resveratrol-3, and resveratrol-4 worked in collaboration to restore the antibiotic effectiveness against two of the seven clinically resistant bacteria. Additionally, either taken alone or in conjunction with levofloxacin at lower dosages, resveratrol and its derivatives, resveratrol-3 and resveratrol-4, greatly decreased the production of biofilms. Also, on soft agar, resveratrol-3 and resveratrol-4 showed a reduction in *Helicobacter pylori* swarming motility [118]. It has been illustrated that trans-resveratrol adjusted the bacterial surface physico-chemical features and increased adhesion and biofilm formation of *Lactobacillus paracasei* [119]. Treating *Limosilactobacillus fermentum* ASBT-2 and *Salmonella enterica* with oxyresveratrol increased the probiotic properties of L. *fermentum* and decreased the growth of *S. enterica* [120]. The efficiency of functioning microspheres in

Table 4
Effect of resveratrol on microbial biofilm and its implications on genitourinary system and skin.

Disorder/Condition	Study design	Doses	Results	Reference
Genitourinary system health				
In vitro				
Urinary tract infection	*E. coli*	10–100 μg/ml	↑*E. coli* killing ↓Biofilm formation, swarming motility, fimbriae production, hemagglutinating ability	[133] 2019
Urogenital tracts infection	*C. albicans*	0–20.32 mM	↓ Biofilm formation, growth, biofilm activities	[134] 2020
Vaginal infection	McCoy cells and *C. trachomatis*	1.5, 3 and 6 μg/ml	↑Antimicrobial and anti-inflammatory effect ↓ *C. trachomatis* propagation, NO production in macrophages	[135] 2020
Urinary catheters infection	*S. epidermidis*	35–130 mg/l	-Inhibited planktonic cells ↓ Metabolic activity of biofilm cells	[136] 2021
Urinary tract infection	*S. saprophyticus* and *P. mirabilis*	114–228 μg/ml	↓ Urease activity, biofilm formation	[137] 2022
Skin health				
In vitro				
Acne vulgaris	*P. acnes*	0.01–0.08 % w/v	↓ Biofilm formation	[143] 2012
Acne vulgaris	Human keratinocyte, monocyte, and *P. acnes*	25–100 μg/ml	↑ Antibacterial activity ↓ Bacterial growth	[144] 2014
Skin infections	*S. aureus* and *S. epidermidis*	50–1000 μg/ml	-Indicated antimicrobial effects ↓ Biofilm activity against *S. aureus*	[145] 2014
Skin infections	THP-1 cells, *S. aureus*	0.018–10 mM	↑ Bacterial membrane leakage, ribosomal protein ↓ Biofilm thickness, chaperone protein downregulation	[146] 2017
Acne vulgaris	*C. acnes*	394 μg/ml	↓ Biofilm formation	[147] 2019
Wound healing	*S. aureus*	2, 4, 8, 16 and 32 μg/ml	↑Fibroblast proliferation and migration ↓ Biofilm formation	[148] 2023

NO: nitric oxide.

facilitating the regulated ciprofloxacin release was evaluated in a different investigation. Resveratrol and D-mannose were used to create these microspheres. When the microspheres' antibacterial activity was assessed, the findings showed that they could successfully prevent *E. coli* from growing [121].

3.4.2. *In vivo studies on digestive system*

The administration of resveratrol to mice with enterocolitis (induced by *C. jejuni*) resulted in ameliorated clinical conditions, and amended function of colonic epithelial barrier. It also reduced colonic epithelial apoptosis and nitric oxide (NO) secretion [122]. It has also been indicated that the growth of *Staphylococcus xylosus* and *Staphylococcus lentus* was inhibited by resveratrol, leading to the improvement of liver fibrosis [123] (Table 3).

In brief, resveratrol reduces the adherence of *S. typhimurium*, *E. coli*, and L. *monocytogenes* to colonic cells *in vitro*, with the former two having the strongest effect. It also has antibacterial properties against *V. cholerae*, *A. butzleri*, *C. coli*, and *C. jejuni*, which destroy biofilms and increase antibiotic efficiency. Resveratrol enhances colonic barrier function, improves enterocolitis outcomes, and inhibits bacterial growth associated with liver fibrosis. While further research is needed to completely understand its mechanisms, these findings emphasize the potential of resveratrol as a treatment for bacterial infections and their consequences (Fig. 3).

3.5. *Effect of resveratrol on genitourinary system*

Microbial biofilms are important in the genitourinary system, especially in the field of infections. Biofilms can form on different tissues in the genitourinary system, including the urethra, bladder, ureters, and kidneys [124,125]. They provide an ideal habitat for pathogens to proliferate and survive, resulting in chronic or recurring infections [126]. Biofilms in the genitourinary tract may have a role in the formation of kidney stones, urinary tract infections, and urinary tract infections linked to catheter use [127,128]. Biofilms allow the microorganisms to adhere to the uroepithelial cells or the surfaces of medical devices like urinary catheters [129]. It is challenging to completely remove the illness once the biofilm has formed since it is naturally resistant to standard antibiotic therapies [130]. Additionally, biofilms can facilitate the exchange of genetic material between microorganisms, promoting the spread of antimicrobial resistance genes and virulence factors [131,132]. Determining the crucial role of microbial biofilms in the genitourinary system is critical to develop effective prevention and treatment methods. Novel techniques to biofilm dispersion, biofilm inhibition, and the formulation of antimicrobial agents with biofilm-specific action are being investigated. It may be possible to develop treatment options that disrupt biofilms and enhance clinical outcomes for genitourinary infections by elucidating the processes involved in biofilm development and persistence.

3.5.1. *In vitro studies on genitourinary system*

In vitro research investigated the effects of ε-viniferin, oxyresveratrol, stilbestrol, suffruticosol A, trans-resveratrol, trans-stilbene, and vitisin A on the generation of biofilm by uropathogenic *E. coli*. The findings demonstrated the effective inhibition of uropathogenic *E. coli* biofilm formation by suffruticosol A, ε-viniferin, oxyresveratrol, vitisin A, and trans-resveratrol. The observation that oxyresveratrol and trans-resveratrol decreased the swarming motility and fimbriae formation in uropathogenic *E. coli* confirmed these outcomes. Moreover, oxyresveratrol and trans-resveratrol significantly decreased the hemagglutinating ability of uropathogenic *E. coli* and elevated the susceptibility of bacteria to being killed by human whole blood [133]. In another study, the effects of micafungin and resveratrol modified molecule were evaluated individually and in combination for their anti-biofilm growth and anti-preformed biofilm activities against *C. albicans*. Micafungin alone showed limited efficacy in completely eradicating fungi. However, when combined with resveratrol modified molecule, synergistic activity was observed, resulting in enhanced anti-growth and anti-preformed biofilm activities [134]. Using free resveratrol and resveratrol liposomes-in-hydrogel against *Chlamydia trachomatis* on McCoy cells resulted in antimicrobial and anti-inflammatory effects and lowered *C. trachomatis* propagation as well as NO production in macrophages [135]. Pterostilbene, a methylated derivative of resveratrol, inhibited the development of planktonic cells while decreasing the metabolic activity of biofilm cells. When compared to standard antibiotics, it demonstrated outstanding efficacy against a clinical isolate. Pterostilbene improved antibiotics (tetracycline and erythromycin) ability to overcome the resistance of *Staphylococcus epidermidis* biofilm cells when administered in combination. Pterostilbene also promoted cytoplasmic membrane permeabilization and surface hydrophobicity alterations in *S. epidermidis* cells [136] (Table 4). Resveratrol could control urease-positive urinary tract pathogens *Proteus mirabilis* and *Staphylococcus saprophyticus* by reducing urease activity and attenuating biofilm formation [137].

To sum up, research shows that natural chemicals such as resveratrol, ε-viniferin, oxyresveratrol, suffruticosol A, t-resveratrol, vitisin A, and pterostilbene can inhibit biofilm development and prevent infections caused by numerous pathogens. These compounds show promise in minimizing biofilm development, modifying bacterial behavior and virulence features, increasing treatment susceptibility, and improving the efficiency of conventional antibiotics. These findings point to the possibility of natural chemicals as adjuvant treatments for biofilm-related disorders.

3.6. Effect of resveratrol on skin

Microbial biofilms have a major effect on skin health and are an important part of the complicated nature of the skin. Although biofilms are frequently linked to infections, especially in wounds, they are also an essential and beneficial element in the skin microbiome [138]. The existence of the biofilm inhibits the colonization of hazardous microbes and promotes a balanced microbial community, acting as a barrier against them. It supports the preservation of the immune system, pH balance, and skin barrier function [139,140]. In contrast, alterations to both the composition and the structure of the biofilm can cause skin disorders including eczema and acne [141,142]. Maintaining ideal skin health and developing targeted treatments for skin disorders need a thorough knowledge of the complicated relationship within the skin microbiota and biofilm.

3.6.1. In vitro studies on skin

It has been reported that exposing *Propionibacterium acnes* to resveratrol resulted in reduced biofilm information [143]. A research was conducted to investigate the therapeutic effects of on acne vulgaris. The results demonstrated that resveratrol had long-lasting antibacterial properties over *P. acnes*. By contrast, benzoyl peroxide, a popular antibacterial acne therapy, exhibited a transient bactericidal effect. Benzoyl peroxide and resveratrol together demonstrated substantial antibacterial activity at first and continued to limit bacterial growth. Using electron microscopy to evaluate *P. acnes* treated with resveratrol showed alterations in bacterial structure, such as an impairment of extracellular fimbrial structures and membrane definition. Furthermore, resveratrol demonstrated lower levels of toxicity compared to benzoyl peroxide [144]. In a study resveratrol was used for prevention and treatment of skin infections (*S. aureus* and *S. epidermidis*). Resveratrol demonstrated antibacterial properties and decreased biofilm activity over *S. aureus*, according to the findings [145]. A group of researchers conducted a study to examine the effectiveness of pterostilbene in contradiction to drug-resistant *S. aureus* and its potential as a treatment for skin bacteria. They found that when tested against clinically isolated vancomycin-intermediate *S. aureus* and methicillin-resistant *S. aureus*, pterostilbene demonstrated better antibacterial properties than resveratrol using the minimum inhibitory concentration assay. Additionally, it was found that pterostilbene decreased the thickness of methicillin-resistant *S. aureus* biofilms. In addition, pterostilbene was readily absorbed by macrophages and showed no toxicity to THP-1 cells, which assisted in eliminating intracellular methicillin-resistant *S. aureus*. The remarkable ability of pterostilbene against methicillin-resistant *S. aureus* was

Table 5
Impact of resveratrol isomers and derivatives on biofilm formation and bacterial behavior.

Compound	Effects	Ref.
ε-viniferin	↓ Biofilm formation and pyocyanin production in *P. aeruginosa* PA14	[93]
	↓ Formation of uropathogenic *E. coli* biofilms	[133]
Glucosyl	↓ Adhesion of *E. coli* and *S. typhimurium* to colonic cells	[107]
Glucosylacyl	↓ Adhesion of *E. coli* and *S. typhimurium* to colonic cells	[107]
Glucuronide	↓ Adhesion of *E. coli* and *S. typhimurium* to colonic cells	[107]
Oxyresveratrol	↓ Formation of *F. nucleatum* biofilms	[70]
	↑ Probiotic properties of L. *fermentum* ↓ Growth of *S. enterica*	[120]
	↓ Formation of uropathogenic *E. coli* biofilms, production of fimbriae and the swarming motility, hemagglutinating ability of uropathogenic *E. coli*	[133]
Piceatannol	↓ Formation of *F. nucleatum* biofilms	[70]
Pterostilbene	↑ Release of cellular contents ↓ Formation of *F. nucleatum* biofilms, bacterial viability	[70]
	↑ Tetracycline and erythromycin ability to overcome the resistance of *S. epidermidis* biofilm cells, cytoplasmic membrane permeabilization and surface hydrophobicity alterations in *S. epidermidis* cells ↓ Development of planktonic cells, metabolic activity of biofilm cells	[136]
	- Stronger antibacterial properties compared to resveratrol against methicillin-resistant *S. aureus* ↑ Bacterial membrane leakage ↓ Thickness of biofilms	[146]
Resveratrol-3	↓ *H. pylori* swarming motility	[118]
Resveratrol-4	↓ *H. pylori* swarming motility	[118]
Suffruticosol A	↓ Formation of uropathogenic *E. coli* biofilms	[133]
Trans-resveratrol	- No significant effect on *E. coli* biofilm formation	[109]
	- Adjusted the physico-chemical properties of the bacterial surface ↑ Adhesion and biofilm formation of L. *paracasei*	[119]
	↓ Formation of uropathogenic *E. coli* biofilms, production of fimbriae and the swarming motility, hemagglutinating ability of uropathogenic *E. coli*	[133]
Viniferin	↓ Formation of *S. pneumoniae* biofilms, membrane integrity	[97]
Vitisin A	↓ Formation of uropathogenic *E. coli* biofilms	[133]

Ref: Reference.

attributed to its ability to upregulate ribosomal proteins, downregulate chaperone proteins, and cause bacterial membrane leakage [146]. It has also been reported that resveratrol could decrease biofilm formation in *Cutibacterium acnes* [147]. Using resveratrol lipid nanoparticles as a wound healing agent increased fibroblast proliferation and migration and decreased biofilm formation of *S. aureus* [148] (Table 4).

In summary, resveratrol and pterostilbene have significant antibacterial properties against bacteria such as *P. acnes* and *S. aureus*. The long-term antibacterial effect of resveratrol inhibits biofilm development by a variety of mechanisms, including changing bacterial structures and causing membrane leakage. It upregulates ribosomal proteins, downregulates chaperone proteins, and ruptures bacterial membranes, increasing its antibacterial activity. Considering these processes and its ability to reduce biofilm, resveratrol appears to be a promising therapy for bacterial infections. More study is needed to properly understand its therapeutic potential, assuring efficacy and safety in clinical settings.

4. Anti-biofilm mechanisms of resveratrol isomers and advanced delivery strategies

Resveratrol exhibits notable anti-biofilm properties across a wide spectrum of bacterial species. Diverse isomers and derivatives of resveratrol have been investigated for their ability to modulate biofilm formation, bacterial adhesion, quorum sensing, and the production of extracellular polymeric substances. These molecular variants can interfere with biofilm integrity by disrupting communication pathways and enhancing susceptibility to antimicrobial agents. Table 5 highlights

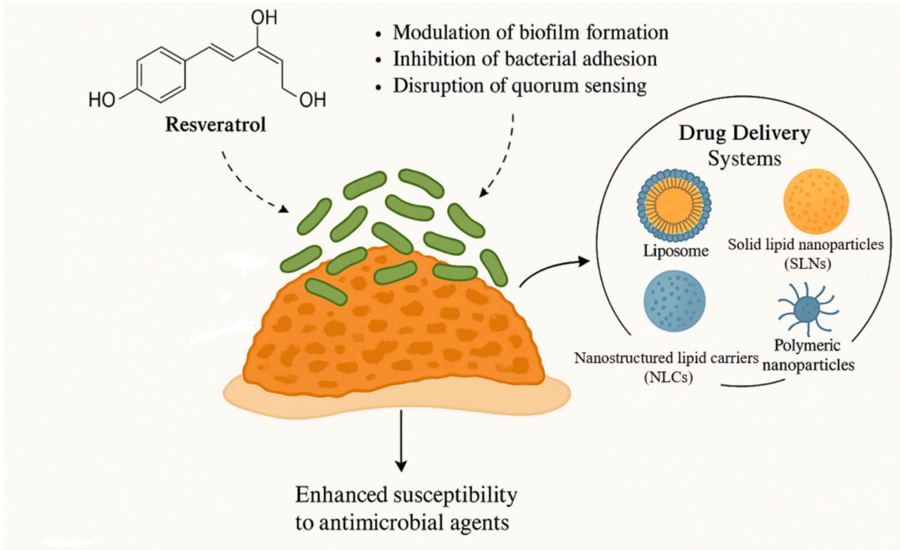

Fig. 4. Mechanistic overview of resveratrol and its derivatives in modulating biofilm architecture, integrated with advanced drug delivery platforms for enhanced therapeutic efficacy.

the effects of different resveratrol compounds on biofilm-related behaviors, underscoring their potential as modulators of microbial colonization.

Despite its promise, resveratrol faces significant challenges in pharmaceutical application due to poor aqueous solubility and limited bioavailability [149]. To overcome these barriers, various strategies have been utilized including synthesis of structural analogs to improve the stability of resveratrol [150,151]. More importantly, innovative drug delivery systems have been developed to maximize its therapeutic potential, particularly in targeting biofilm-related infections. Lipid-based formulations—including liposomes, solid lipid nanoparticles (SLNs), and nanostructured lipid carriers (NLCs)—enhance solubility and enable targeted delivery to biofilm niches [152–154]. Similarly, polymeric nanoparticles such as micelles, dendrimers, and polymer-drug conjugates protect resveratrol from degradation while improving release kinetics and cellular uptake [155,156]. Other promising platforms include nanoemulsion systems, which optimize absorption and allow site-specific delivery [157–159]; cyclodextrin-based inclusion complexes, which improve water solubility and chemical stability [160]; and solid dispersions, which increase dissolution rates by enhancing surface contact [161,162]. Collectively, these technologies offer a multifaceted approach to resveratrol administration, potentially amplifying its efficacy against resilient biofilm structures.

To conceptually illustrate these interactions, a hypothetical figure (Fig. 4) has been proposed that delineates the synergy between resveratrol's chemical diversity and advanced delivery vehicles in disrupting biofilm architecture. This integrative model highlights future perspectives for designing next-generation anti-biofilm therapies that leverage both compound selection and formulation strategy.

5. Future perspectives

While the manuscript provides compelling evidence on the potential value of resveratrol in disrupting microbial biofilms, there are several avenues for future research that can further enhance our understanding and application of this compound. Future research should focus on optimizing dosage and delivery methods, assessing long-term effects and safety, conducting well-designed clinical trials, exploring combination therapies, investigating underlying mechanisms, and expanding the translational applications of resveratrol. These efforts will contribute to the development of effective and innovative antimicrobial therapies targeting biofilms and bacterial infections.

6. Conclusion

In conclusion, the manuscript provides comprehensive evidence supporting the potential advantages of resveratrol in disrupting microbial biofilms in the human body. Resveratrol exhibits remarkable antimicrobial and anti-biofilm properties, primarily through its anti-inflammatory effects, which involve modulating signaling pathways and inhibiting the expression of inflammatory factors. Moreover, resveratrol effectively inhibits the formation of biofilm, growth, and biofilm-related genes expression. It further disrupts bacterial cell membranes and reduces bacterial viability. Additionally, resveratrol demonstrates the ability to inhibit the expression of virulence genes and protease activity, thereby attenuating the pathogenic properties of bacteria. In addition, resveratrol shows antibacterial effects against specific pathogens, reducing acid production, inhibiting polysaccharide synthesis, and compromising biofilm formation. Furthermore, resveratrol displays significant anti-quorum sensing activity, enhancing the antimicrobial properties of antibiotics and disrupting biofilm architecture by inhibiting quorum sensing gene expression. To optimize resveratrol's dosage, long-term effects, and delivery systems for a range of medical diseases, more research is needed. These findings provide valuable insights and pave the way for the development of innovative antimicrobial therapies using resveratrol and its formulations.

CRediT authorship contribution statement

Mahboobeh Ghasemzadeh Rahbardar: Writing – original draft. **Prashant Kesharwani:** Writing – review & editing, Supervision, Conceptualization. **Amirhossein Sahebkar:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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